

DATE: TUESDAY 8TH DECEMBER 2020

TIME: 11.00 A.M. – 1.00 P.M.

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any two other questions

Useful Information

$$R = 8.314 \text{ J mole}^{-1} \text{ K}^{-1} \text{ or } = 0.08206 \text{ litre atm mole}^{-1} \text{ K}^{-1}$$

$$1 \text{ atm pressure} = 760 \text{ mm Hg or } = 101325 \text{ N m}^{-2}$$

$$\text{Molar gas volume at STP } 22.4 \text{ litres}$$

$$F = 96500 \text{ C mol}^{-1}$$

Standard Reduction potential table is provided

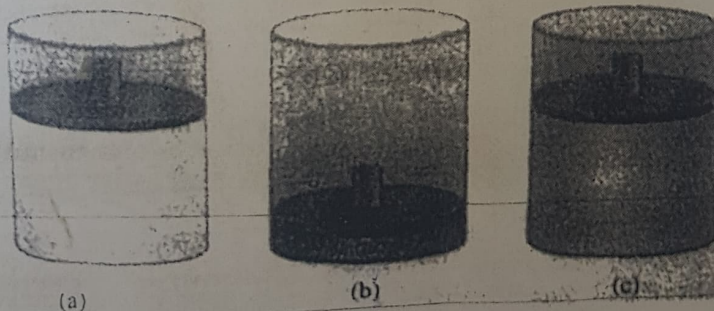
QUESTION ONE (30 marks)

- a) Explain why mercury is a more suitable substance to use in a barometer than water (2 marks)
- b) Consider the following gaseous sample in a cylinder fitted with a movable piston. Initially there are n moles of the gas at temperature T , pressure P and volume V (fig 1)



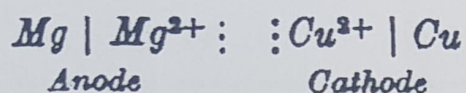
Fig 1.

Choose the cylinder that correctly represents the gas after each of the following changes and explain your answer.



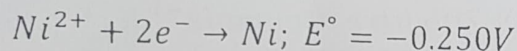
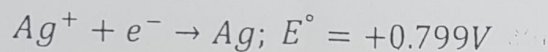
- (i) The pressure on the piston is tripled at constant n and T (2 marks)
- (ii) The temperature is doubled at constant n and P (2 marks)
- (iii) n moles of another gas are added at constant T and P . (2 marks)
- (iv) T is halved and pressure on the piston is reduced to a quarter of its original value. (2 marks)

c) (i) What is a cell diagram? Write the half-cell reactions for the following cell:



(3 marks)

(ii) Write the cell reaction and calculate the cell e.m.f. from the data: (6 marks)



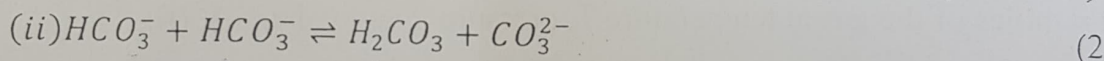
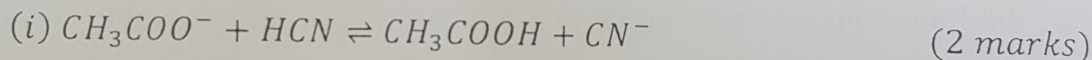
d) Describe the effect on pH (increase, decrease or no change) that results from each of the following additions:

(i) potassium acetate to an acetic acid solution (1 mark)

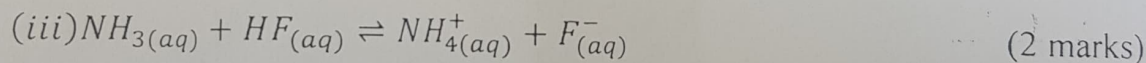
(ii) ammonium nitrate to an ammonia solution (1 mark)

(iii) sodium formate (HCOONa) to a formic acid (HCOOH) solution (1 mark)

e) Identify the acid base conjugate pairs for the following reactions:



(iii)



f) What is the rule for writing the equilibrium constant for the overall reaction involving two or more reactions (2 marks)

QUESTION TWO (20 marks)

a) Define the following terms:

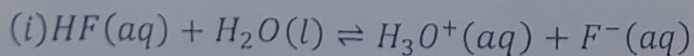
(3 marks)

(i) Buffer solution

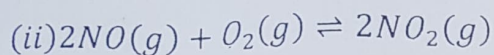
(ii) Law of mass action

(iii) Homogeneous equilibrium

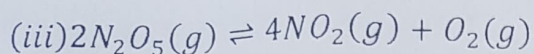
b) Write the expressions for K_c and K_p if applicable for each of the following reversible reactions at equilibrium:



(2 marks)



(2 marks)



(2 marks)

c) The equilibrium constant K_p for the decomposition of phosphorous pentachloride to phosphorous trichloride and molecular chlorine is found to be 1.05 at 250°C. If the equilibrium partial pressure of PCl_5 and PCl_3 are 0.875 atm and 0.463 atm respectively, what is the equilibrium partial pressure of Cl_2 at 250°C. (4 marks)

d) Calculate the pH of a 0.036M nitrous acid (HNO_2) solution; $K_a = 4.5 \times 10^{-4}$ (7 marks)

QUESTION THREE (20 marks)

a) Define the following terms:

i) e.m.f (1 mark)

ii) voltaic cell (1 mark)

iii) SHE electrode (1 mark)

b) Outline the guidelines for writing equilibrium constant expressions (5 marks)

c) The pH of a 0.10M solution of formic acid ($HCOOH$) is 2.39. What is the K_a of the acid. (6 marks)

d) Which of the following solutions can be classified as buffer systems? Explain your answer. (6 marks)

(i) $NaClO_4/HClO_4$ (ii) C_5H_5N/C_5H_5NHCl (iii) $Na_2CO_3/NaHCO_3$ (iv) KF/HF

QUESTION FOUR (20 marks)

a) Define the following terms:

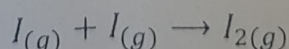
i) Solubility product (1 mark)

ii) Molar solubility (1 mark)

iii) Ion-product constant (1 mark)

iv) Chemical kinetics (1 mark)

b) Iodine atoms combines to form molecular iodine in the gas phase



This reaction follows second-order kinetics and has the high constant $7.0 \times 10^9 / M.s$ at 23°C

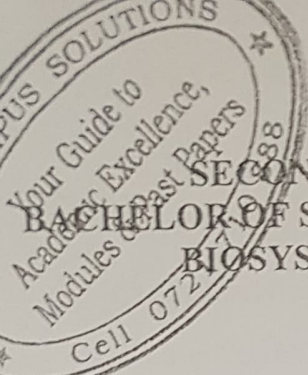
(i) If the initial concentration of I (iodine) was 0.086M, calculate the concentration after 2.0min. (3 marks)

$$Kt = \frac{1}{[A]_0} - \frac{1}{[A]}$$

- (ii) Calculate the half-life of the reaction if the initial concentration of I is $0.60M$ and if it is $0.42M$. (4 marks)
- c) Calculate the pH of: $k_t = n_f$
- (i) a $1.0 \times 10^{-3} M$ HCl solution (4 marks)
- (ii) A $0.020M$ $Ba(OH)_2$ solution (5 marks)

QUESTION FIVE (20 marks)

- a) State the following laws:
- (i) Rate Law (1 mark)
- (ii) Boyle's law (1 mark)
- b) Calculate the solubility of copper silver chloride ($AgCl$) in g/L , $K_{sp} = 1.6 \times 10^{-10}$ (4 marks)
- c) Classify each of the species as a Brønsted acid or base or both
- (i) H_2O (ii) OH^- (iii) CO_3^{2-} (3 marks)
- d) (i) Calculate the pH of a buffer system containing $1.0M$ CH_3COOH and $1.0M$ CH_3COONa (5 marks)
- (ii) What is the pH of the buffer system after the addition of 0.10 mole of gaseous HCl to $1.0L$ of the solution? Assume that the volume of the solution does not change when the HCl is added. (K_a of CH_3COOH is 1.8×10^{-5}) (6 marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2018/2019

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (PETROLEUM, ELECTRICAL, MECHANICAL, CIVIL,
BIOSYSTEMS, BIOMEDICAL & AEROSPACE ENGINEERING)

ECU 102: CHEMISTRY FOR ENGINEERS II

DATE: MONDAY 8TH APRIL, 2019

TIME: 2.00 P.M - 5.00 P.M

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any two other questions

Useful Information

$$R = 8.314 \text{ J mole}^{-1} \text{ K}^{-1} \text{ or } = 0.08206 \text{ litre atm mole}^{-1} \text{ K}^{-1}$$

$$1 \text{ atm pressure} = 760 \text{ mm Hg or } = 101325 \text{ N m}^{-2}$$

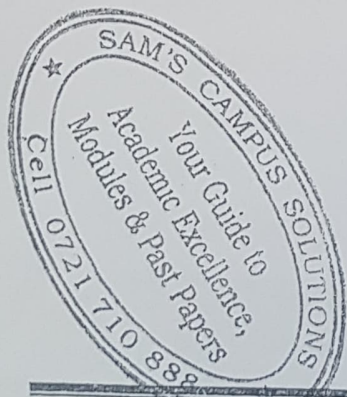
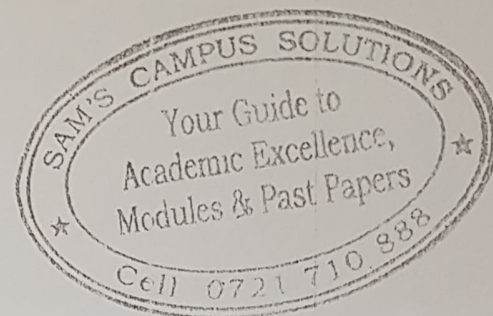
Molar gas volume at STP 22.4 litres

$$F = 96500 \text{ C mol}^{-1}$$

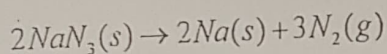
Standard Reduction potential table is provided

QUESTION ONE (30 marks)

- a) i) State Le Châtelier's principle (1 marks)
ii) List two factors that affect chemical equilibrium (2 marks)
- b) Briefly describe each of the following idea or phenomena:
 - i) Dynamic equilibrium (1 marks)
 - ii) Physical equilibrium (1 marks)
 - iii) Chemical equilibrium (1 marks)
- c) Explain the important distinction between each pair of terms:
 - i) Homogeneous and heterogeneous equilibrium (1 marks)
 - ii) K_c and K_p (1 marks)
- d) i) State Dalton's law of partial pressures. [1 marks]
ii) 4.0 g of oxygen (O_2) and 7.0 g of nitrogen (N_2) gases were put in a container of volume of 5.0 litres at 75 °C. Determine:
 - i) Partial pressure of O_2 gas. (2 marks)
 - ii) Partial pressure of N_2 gas. (2 marks)
 - iii) Total pressure exerted by the two gases. (2 marks)



e) Sodium azide (NaN_3) is used in some automobile air gas. The impact of a collision triggers the decomposition of NaN_3 as follows:



The nitrogen gas produced quickly inflates the bag between the driver and the windshield and dash board. Calculate the volume (in litres) of N_2 generated at 80°C and 823 mmHg by the decomposition of 60.0g of NaN_3 (5 marks)

f) The best known triprotic acid is phosphoric acid. Write its ionization equations (3 marks)

g) Classify each of the following species in aqueous solution as a Bronsted acid or base:

i) NO_2^- ii) HCO_3^- (2 marks)

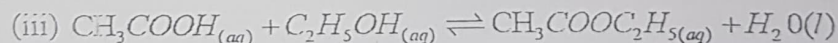
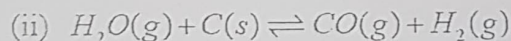
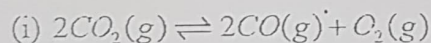
h) What is the standard emf of a galvanic cell made of a Cd electrode in a 1.0M $\text{Cd}(\text{NO}_3)_2$ solution and a Cr electrode in a 1.0M $\text{Cr}(\text{NO}_3)_3$ solution at 25°C ? (5 marks)

QUESTION TWO (20 marks)

a) Define the following terms:

- (i) Conjugate base and conjugate acid
- (ii) pK_a for a weak acid
- (iii) Triprotic acid

b) Write the expressions for K_c and K_p if applicable for each of the following reversible reactions at equilibrium: (4marks)



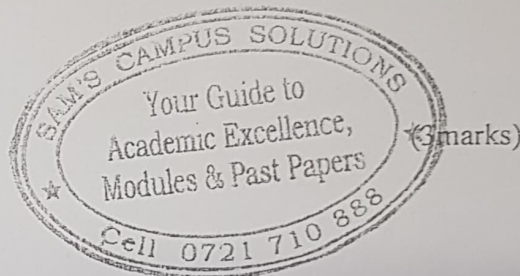
c) List any four general properties of bases (2 marks)

d) The concentration of OH^- ions in a certain household ammonia cleaning solution is 0.0025M. Calculate the concentration of H^+ ions. (4 marks)

e) Which of the following solutions can act as a buffer or not?

(i) KCl/HCl (ii) $\text{KHSO}_4/\text{H}_2\text{SO}_4$ (iii) $\text{KNO}_2/\text{HNO}_2$ (3 marks)

f) The pH of a 0.060M weak monoprotic acid is 3.44. Calculate the K_a of the acid? (4 marks)



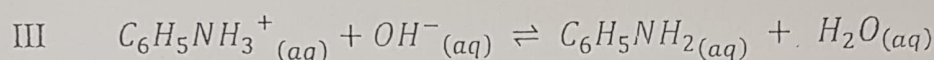
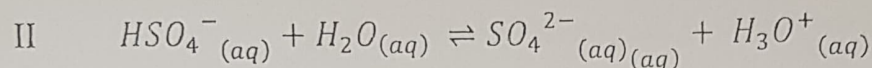
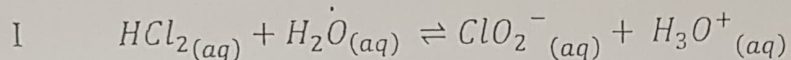
*KOH + HCl
KCl + H2O*

QUESTION THREE (20 marks)

a) Define the following terms:

- i) Standard reduction potential (1 mark)
- ii) Cell voltage (1 mark)
- iii) Cathode (1 mark)

b) For each of the following identify the acid-base conjugate pairs (3 marks)



c) Equilibrium involving $\text{SO}_{2(\text{g})}$, $\text{O}_{2(\text{g})}$ and $\text{SO}_{3(\text{g})}$ is important in sulphuric acid production.

When a 0.0200 mol sample of $\text{SO}_{3(\text{g})}$ is introduced in an evacuated 1.52 L

vessel at 627 °C, 0.0142 mole $\text{SO}_{3(\text{g})}$ is found to be present at equilibrium. Calculate:

- i) $[\text{SO}_2]$ (2 mark)
- ii) $[\text{O}_2]$ (2mark)
- iii) K_c (2 mark)
- iii) K_p (2 marks)

d) The solubility product of PbI_2 at 25 °C $K_{sp} = 7.1 \times 10^{-9} \text{ mol}^3 \text{ litre}^{-3}$.

Calculate the molar solubility PbI_2 of at these conditions. (2 marks)

e) i) Calculate the pH of 0.00250 M HNO_2 acid, given $K_{a(\text{HNO}_2)}$ is 7.2×10^{-4}

(2 marks)

ii) With a specific example, define a buffer solution (2 marks)

QUESTION FOUR (20 marks)

a) i) List four the characteristics of an ideal gas. (4 marks)

ii) Write the ideal gas equation and give the units for each term in the equation. (2 marks)

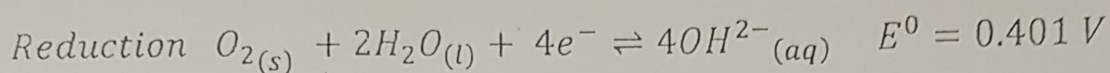
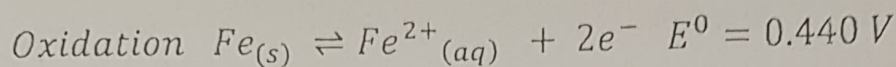
b) A small bubble rises from the bottom of a lake, where the temperature and pressure are 8°C and 6.4 atm, to the water's surface, where the temperature is 25°C and the pressure is 1.0 atm. Calculate the final volume (in ml) of the bubble if its initial volume was 2.1ml. (5 marks)

c) Calculate the pH of a 0.40M ammonia solution? Show how the concentrations of the pertinent species change. $K_b = 1.8 \times 10^{-5}$ (7 marks)

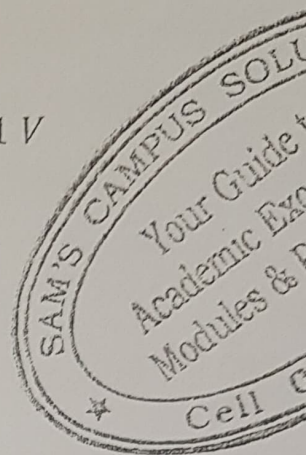
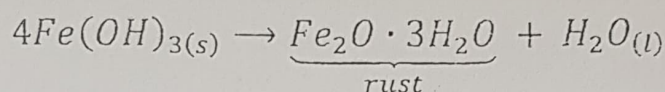
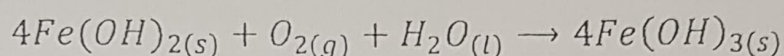
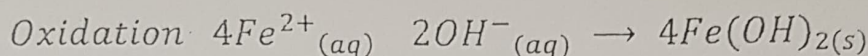
d) Write the equation relating K_c and K_p and define all the terms (2 marks)

QUESTION FIVE (20 marks)

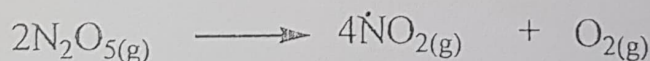
- a) Corrosion is unwanted voltaic cells reaction. The two simple half-equation of electrolytic corrosions are:



Then other reactions

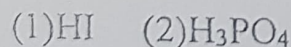


- i) Calculate the e.m.f. of the electrolytic corrosions (2 marks)
- ii) Is rusting spontaneous reaction? Give reasons (2 marks)
- iii) Determine the equilibrium constant of the electrolytic corrosions at 25 °C (2 marks)
- iv) State three methods of protecting iron from rusting (3 marks)
- v) State economic effect of rusting (2marks)
- b) The decomposition of dinitrogen pentoxide is a first-order reaction with rate constant of $5.1 \times 10^{-4} \text{ s}^{-1}$ at 45 °C.



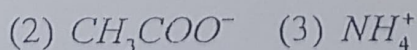
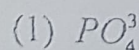
- i) If the initial concentration of N_2O_5 was 0.25 M, calculate the concentration after 3.2 minutes. (2 marks)
- ii) How long will it take for the concentration of N_2O_5 to decrease from 0.25 M to 0.15 M. (2 marks)

- c) (i) Identify the following as weak or strong acid



(2 marks)

- (ii) Identify each of the species as a Brønsted acid, base or both:



(3 marks)

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (PETROLEUM, ELECTRICAL, MECHANICAL, CIVIL, BIOSYSTEMS,
BIOMEDICAL & AEROSPACE ENGINEERING)

ECU 102: CHEMISTRY FOR ENGINEERING II

DATE: Tuesday 21st August 2018 -

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any other TWO questions
3. Standard Reduction potential table is provided

Useful information:

$$R=8.314\text{J.mol}^{-1}.\text{k}^{-1}=0.08205\text{L.atm.K}^{-1}.\text{mol}^{-1}$$

$$1\text{ atm}=760\text{mmHg}=101,325\text{ Pa}$$

$$F=96500\text{Cmol}^{-1}$$



QUESTION ONE (30 marks)

- a) Name FIVE elements that exist as gases (5marks)
- b) State FOUR postulates that the kinetic molecular theory is based on (4marks)
- c) Define pH. Why do chemists normally choose to discuss the acidity of a solution in terms of pH rather than hydrogen ion concentration, $[\text{H}^+]$ (5 marks)
- d) Which of the following compounds will be more soluble in acidic solution than in water? Explain your answer. (6 marks)
 - i) CuS
 - ii) AgCl
 - iii) PbSO_4

QUESTION TWO (20 marks)

(4 marks)

a) State four functions of an Ideal gas

b) A mixture of gases contains 4.46 moles of Neon, 0.74 mole of argon and 2.15 moles of xenon. Calculate the partial pressures of the gases if the total pressure is 2.00 atm at a certain temperature

(7 marks)

c) Chemical analysis of a gaseous compound showed that it contained 33.0% silicon (Si) and 67.0% fluorine (F) by mass. At 35°C, 0.21L of the compound exerted a pressure of 1.70 atm. If the mass of 0.21L of the compound is 2.38g, calculate the molar mass of the compound. $M(\text{Si}) = 28.09\text{g}$, $M(\text{F}) = 19.0\text{g}$

(9 marks)

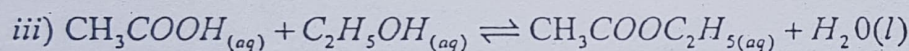
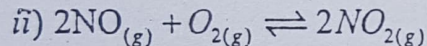
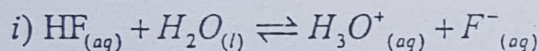
QUESTIONS THREE (20 marks)

a) List the factors that affect Chemical equilibrium

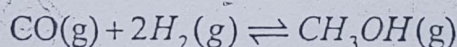
(3 marks)

b) Write the expressions for K_c and K_p if applicable for each of the following reversible reactions at equilibrium:

(6 marks)



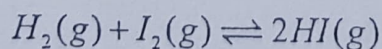
c) i) Methanol is manufactured industrially by the reaction below:



The equilibrium constant (K_c) for the reaction is 10.5 at 220°C. What is the value of K_p at this temperature

(5 marks)

ii) A mixture of 0.500 mol H_2 and 0.500 mol I_2 was placed in a 1.00L stainless-steel flask at 430°C. The equilibrium constant K_c for the reaction below is 54.3 at this temperature. Calculate the concentrations of H_2 , I_2 and HI at equilibrium



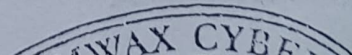
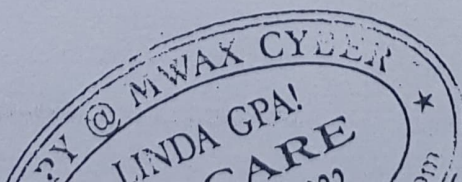
(6 marks)

QUESTION FOUR (20 marks)

- a) List any THREE general properties of acids (3 marks)
- b) (i) Give Arrhenius's and Brønsted's definitions of an acid and a base. (4 marks)
- (ii) Why are the Brønsted's definitions more useful in describing acid-base properties? (1 mark)
- c) With the aid of a reaction in each case; describe the following acids: (2 marks)
- (i) Monoprotic
- (ii) Triprotic *acetic acid* (2 marks)
- d) i) The pH of rainwater collected in Industrial area on a particular day was 4.82. Calculate the H^+ ion concentration of the rainwater. (4 marks)
- ii) ~~Calculate~~ the pH of 0.10M solution of acid ($HCOOH$) is 2.39. What is the K_a of the acid? (4 marks)

QUESTION FIVE (20 marks)

- a) Define the following terms: (3 marks)
- (i) Electromotive force
- (ii) Anode
- (iii) Galvanic cell
- b) What is the function of a salt bridge (2 marks)
- c) Calculate the equilibrium constant for the following reaction at $25^\circ C$ (7 marks)
- $$Sn(s) + 2Cu^{2+}(aq) \rightleftharpoons Sn^{2+}(aq) + 2Cu^+(aq) \quad K =$$
- d) (i) One half of an electrochemical cell consists of a pure nickel electrode in a solution of Ni^{2+} ions; the other half is a cadmium electrode immersed in a Cd^{2+} solution. $E^\circ_{Cd} = -0.403V$, $E^\circ_{Ni} = -0.250V$.
- If the cell is a standard one, write the spontaneous overall reaction and calculate the voltage that is generated (4 marks)
- (ii) Compute the cell potential at $25^\circ C$ if the Cd^{2+} and Ni^{2+} concentrations are 0.5 and $10^{-3}M$, respectively. Is the spontaneous direction still the same as for the standard cell? (4 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (PETROLEUM ENGINEERING),
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING),
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING),
BACHELOR OF SCIENCE (CIVIL ENGINEERING),
BACHELOR OF SCIENCE (BIOSYSTEM ENGINEERING), BACHELOR OF SCIENCE
(BIOMEDICAL ENGINEERING) AND
BACHELOR OF SCIENCE (AEROSPACE ENGINEERING)

ECU 102: CHEMISTRY FOR ENGINEERS II

DATE: Thursday, 15th June 2017

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any two other questions

Useful information

1 atm = 760 mm Hg = 101,325 Pa

Standard Reduction potential table is provided

QUESTION ONE (30 MARKS)

(a) State the following gas laws in words and also in the form of an equation. In each case indicate the conditions under which the laws are applicable:

- (i) Charles's law (4 marks)
- (ii) Avogadro's law (4 marks)

(b) Explain the difference between physical equilibrium and chemical equilibrium. Give two examples of each. (8 marks)

(c) Define Bronsted acids and bases. Give an example of a conjugate pair. (5 marks)

(d) Which of the following compounds will be more soluble in acidic solution than in water?

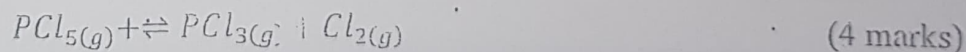
- (i) CuS
- (ii) AgCl
- (iii) PbSO₄ (9 marks)

QUESTION TWO (20 MARKS)

- (a) List the physical characteristics of gases and name two compounds that exist as gases at room temperature. (5 marks)
- (b) 3.5 moles of NH_3 occupy 5.20L at 47°C , calculate the pressure of gas (in atm) using:
- (i) The ideal gas equation (4 marks)
 - (ii) The van der Waals equation (4 marks)
- (c) 2 glass bulbs of equal volume are connected by a narrow tube and are filled with a gas at 273.15K and 1 bar. One of the bulbs is placed in melting ice and the other in a water bath at 62°C . What is the new value of P inside the bulb? (7 marks)

QUESTION THREE (20 MARKS)

- (a) Discuss briefly how the following factors affect Chemical equilibrium;
- (i) Changes in temperature (3 marks)
 - (ii) Changes in Volume and Pressure (3 marks)
- (b) Write the equilibrium constant expression K_c and K_p if applicable for each of the following heterogeneous systems:
- (i) $(\text{NH}_4)_2\text{Se}_{(s)} \rightleftharpoons 2\text{NH}_{3(g)} + \text{H}_2\text{Se}_{(g)}$
 - (ii) $\text{AgCl}_{(s)} \rightleftharpoons \text{Ag}_{(aq)}^+ + \text{Cl}_{(aq)}^-$
 - (iii) $\text{P}_{4(s)} + 6\text{Cl}_{2(g)} \rightleftharpoons 4\text{PCl}_{3(l)}$
- (c) (i) The equilibrium constant K_p for the decomposition of phosphorous pentachloride to phosphorous trichloride and molecular chlorine is found to be 1.05 at 250°C . If the equilibrium partial pressures of PCl_5 and PCl_3 are 0.875 atm and 0.463 atm, respectively, compute the equilibrium partial pressure of Cl_2 at 250°C .



- (iii) A mixture of SO_2 and O_2 in the molar ratio 2:1 was kept over platinum catalyst in a reaction vessel at 250°C . Total pressure at equilibrium was found to be 10 atm. If 60% SO_2 can be converted to SO_3 , what is the K_p for the reaction?

(5 marks)

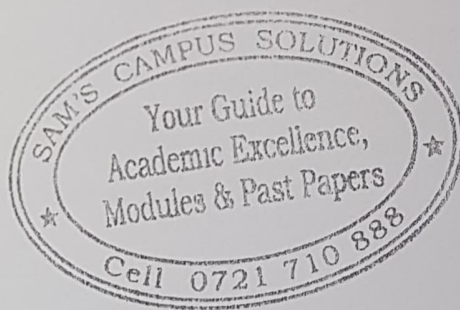
QUESTION FOUR (20 MARKS)

- (a) In order for a species to act as Bronsted base, an atom in the species must possess a lone pair of electrons. Explain why. (2 marks)
- (b) Predict the direction of the following reactions in aqueous solution:
- (i) $\text{HNO}_2(\text{aq}) + \text{CN}^-(\text{aq}) \rightleftharpoons \text{HCN}(\text{aq}) + \text{NO}_2^-(\text{aq})$
- (ii) $\text{CH}_3\text{COOH}(\text{aq}) + \text{HCOO}^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{HCOOH}(\text{aq})$ (4 marks)
- (c) Identify the acid-base conjugate pairs in each of the following reactions:
- (i) $\text{NH}_3(\text{aq}) + \text{HF}(\text{aq}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{F}^-(\text{aq})$
- (ii) $\text{H}_2\text{PO}_4^- + \text{NH}_3 \rightleftharpoons \text{HPO}_4^{2-} + \text{NH}_4^+$
- (iii) $\text{CH}_3\text{COO}^- + \text{HCN} \rightleftharpoons \text{CH}_3\text{COOH} + \text{CN}^-$ (6 marks)
- (d) (i) The pH of rainwater collected in Industrial area on a particular day was 4.82. Calculate the H^+ ion concentration of the rainwater. (4 marks)
- (ii) Calculate the pH of 0.020M $\text{Ba}(\text{OH})_2$ solution. (4 marks)

QUESTION FIVE (20 MARKS)

- (a) Define the following terms:

- (i) Standard reduction potential
- (ii) Cell voltage
- (iii) Cathode



(3 marks)

- (b) Describe the basic features of a galvanic cell. Why are the two components of the cell separated from each other? (4 marks)
- (c) (i) Predict what will happen if molecular bromine (Br_2) is added to a solution containing NaCl and NaI at 25°C . Assume all species are in their standard states. (4 marks)
- (ii) A galvanic cell consists of Mg electrode in a $1.0 \text{ M Mg}(\text{NO}_3)_2$ solution and a Ag electrode in 1.0 M AgNO_3 solution. Calculate the standard emf of this cell at 25°C . (4 marks)
- (d) The pH of a 0.10 M solution of formic acid (HCOOH) is 2.39. What is the K_a of the acid? (5 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2011/2012

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN ENGINEERING

ECU 102: CHEMISTRY FOR ENGINEERS II

DATE: TUESDAY, 10TH APRIL 2012

TIME: 11.00 A.M. – 1.00 P.M.

Instructions:

Answer all questions

Useful information

$$R=8.314 \text{ JK}^{-1}\text{mol}^{-1}=0.082051 \text{ atm K}^{-1}\text{mol}^{-1}$$

$$0^{\circ}\text{C}=273\text{K}, N=14, O=16, S=32, F=19$$

$$C=3.0 \times 10^8 \text{ m/s}$$

$$a=3.59 \text{ L}^2 \text{ atm mol}^{-2};$$

$$b=0.0427 \text{ L mol}^{-1}$$

Question 1. [Total 12 Marks]

- a. Derive the ideal gas equation, $PV=nRT$. [4 Marks]
- b. Calculate the pressure exerted by 1.00 mol of CO_2 when it is confined to a volume of 2.50 L at 298 K by using.
 - i) The ideal gas equation. [3 Marks]
 - ii) The Vander Waals' equation [3 Marks]
 - iii) On comparing the two results, give an explanation for the difference between them. [2 Marks]

containing SF_6 at STP. How do these systems compare with respect to

- i) Number of molecules. [3 Marks]
- ii) Density of gases.
- iii) Average kinetic energy of molecules.
- iv) Rate of diffusion through a pinhole Leak. [4 Marks]

b) A measured amount of Argon gas escaped through a tiny hole in 77.3 Sec.

A similar amount of unknown gas escaped under the same condition in 97.6 Sec. Calculate the molar mass of the unknown given that the molar mass of Argon gas is 39.95g/mol [4 Marks]

c) Oxygen used in respiration therapy is stored at room temperature under a pressure of 1.50×10^2 atm in gas cylinder with a volume of 60.0 L.

- i) What volume would the gas in one of these cylinder occupy at a pressure of 750mmHg. [3 Marks]
- ii) If the oxygen flows to the patient adjusted to 8.00L per minute at room temperature and 750mmHg, How long will the tank of gas last. (room temperature = 25°C) [3 Marks]

Question 2. [Total 14 Marks]

a. Suppose you have two 1-L flasks one containing N_2 at STP, and the other containing SF_6 at STP. How do these systems compare with respect to

- i) Number of molecules. [3 Marks]
- ii) Density of gases.
- iii) Average kinetic energy of molecules.
- iv) Rate of diffusion through a pinhole Leak. [4 Marks]

b) A measured amount of Argon gas escaped through a tiny hole in 77.3 Sec.

A similar amount of unknown gas escaped under the same condition in 97.6 Sec. Calculate the molar mass of the unknown given that the molar mass of Argon gas is 39.95g/mol [4 Marks]

c) Oxygen used in respiration therapy is stored at room temperature under a pressure of 1.50×10^2 atm in gas cylinder with a volume of 60.0 L.

- i) What volume would the gas in one of these cylinder occupy at a pressure of 750mmHg. [3 Marks]
- ii) If the oxygen flows to the patient adjusted to 8.00L per minute at room temperature and 750mmHg, How long will the tank of gas last .(room temperature= 25^0 c) [3 Marks]

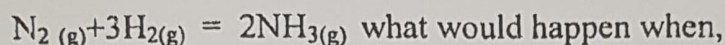
Question 3. [Total 12 Marks]

a) A sample of COCl_2 gas is introduced into a constant volume vessel at 395°C and observed to exert an initial pressure of 0.351 atm. Calculate

i) The partial pressure of each gas.. [3 Marks]

ii) The total gas pressure [1 Mark]

b) For the following reaction



i) Increasing the volume of container in which the reaction is occurring [1 mark]

ii) Increasing the pressure at equilibrium [1 Mark]

iii) Decreasing the pressure at equilibrium [1 Mark]

iv) Derive the relationship between K_C and K_p for the same reaction [5 Mark]

Question 4. [Total 22 Marks]

a) i) Write the equilibrium expression for the following reaction.



ii) for the same reaction, calculate the concentration of H_2 when

$[\text{H}] = 0.5\text{M}$, $[\text{I}] = 0.5\text{M}$ and $[\text{I}] = 0.63\text{M}$ and $K_c = 0.10$ [4 Marks]

b) What will be the pH of

i) A solution of 0.00025M , HCL

ii) A solution of 0.055M , NaOH .

iii) A solution of 0.015M , $\text{Ba}(\text{OH})_2$ [3 Marks].

c) Write the overall cell reaction for the voltaic cell,

i. $\text{Ti}_{(s)}/\text{Ti}_{(aq)}/\text{Sn}^{2+}_{(aq)}/\text{Sn}_{(s)}$ [4 Marks]

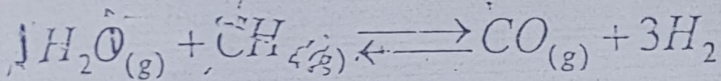
ii. How long will it take to deposit 4 g of tin in the above cell when a current of 0.240 A flows through it. [4 Marks]

iii. During the electrolysis of 1 M H^+ ions using Zinc electrodes.

What would you expect to be released at the cathode electrode .
[2 Marks]

iv. Experimentally what do you expect to be simultaneously deposited and why. [3 Mark]

THE END



- i) Determine the value of K_c at this temperature (4 marks)
- ii) Explain the effect of (pressure) changes on the position of equilibrium of the reaction (3 marks)

QUESTION FOUR (TOTAL 18 MARKS)

a) What will be the pH of...

- i) A solution of Hydrochloric acid of concentration of $0.001 \text{ mol dm}^{-3}$ (2 marks)
- ii) A solution of Butanoic Acid $C_3H_7CO_2H$ for which $K_a = 1.51 \times 10^{-5} \text{ mol dm}^{-3}$

(4 marks)

b) For the chemical cell :

- i) A voltaic cell is constructed with an Sn/Sn^{2+} half cell and a Zn/Zn^{2+} half cell.

Measurements show that the zinc ^(+ve) Electrode is Negative.

i) Write the balanced half-reactions

(2 marks)

ii) Write the overall spontaneous reaction in the cell

(2 marks)

iii) Diagram the cell and label the electrodes with their charges

(4 marks)

iv) Show the direction of electron flow in the circuit and of Cation and Anion flow in the salt Bridge.

(4 marks)

END

